

EX.NO: 5

Stack based buffer overflow attacks

ROLL NO : 231901026

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Sudo Buffer Overflow

Exploring CVE-2019-18634 in the Unix sudo program. Room two in the SudoVulns series

Buffer Overflow Prep

Practice stack based buffer overflows!

The image displays two screenshots of a cybersecurity challenge platform. The top screenshot shows the 'Sudo Buffer Overflow' challenge interface. It features a dark theme with a green header bar. The title 'Sudo Buffer Overflow' is prominently displayed, followed by a subtitle: 'A tutorial room exploring CVE-2019-18634 in the Unix Sudo Program. Room Two in the SudoVulns Series'. Below the title, there are buttons for 'Share your achievement', 'Start AttackBox', 'Help', 'Save Room', and 'Options'. A progress bar indicates 'Room completed (100%)'. The task list shows 'Task 1: Deploy' and 'Task 2: Buffer Overflow'. A recommendation section asks 'How likely are you to recommend this room to others?' with a scale from 1 to 10 and a 'Submit now' button.

The bottom screenshot shows the 'Buffer Overflow Prep' challenge interface. It also has a dark theme with a green header bar. The title 'Buffer Overflow Prep' is displayed, followed by the subtitle 'Practice stack based buffer overflows!'. Below the title, there are buttons for 'Share your achievement', 'Start AttackBox', 'Help', 'Save Room', and 'Options'. A progress bar indicates 'Room completed (100%)'. A scoreboard section shows a chart with multiple colored lines representing different players' progress. The chart has a y-axis ranging from 0 to 100 and an x-axis with various colored lines representing different players' progress.

TASK 2 : BUFFER OVERFLOW

Use the pre-compiled exploit in the VM to get a root shell.

No answer needed

✓ Correct Answer

What's the flag in /root/root.txt?

THM{buff3r\_0v3rfl0w\_rul3s}

✓ Correct Answer

## Buffer Overflow Prep

### TASK 2 oscp.exe - OVERFLOW1

What is the EIP offset for OVERFLOW1?

1978

✓ Correct Answer

In byte order (e.g. \x00\x01\x02) and including the null byte \x00, what were the badchars for OVERFLOW1?

\x00\x07\x2e\xa0

✓ Correct Answer

💡 Hint

### TASK 3 : oscp.exe - OVERFLOW2

What is the EIP offset for OVERFLOW2?

634

✓ Correct Answer

In byte order (e.g. \x00\x01\x02) and including the null byte \x00, what were the badchars for OVERFLOW2?

\x00\x23\x3c\x83\xba

✓ Correct Answer

### TASK 4 oscp.exe – OVERFLOW3

What is the EIP offset for OVERFLOW3?

1274

✓ Correct Answer

In byte order (e.g. \x00\x01\x02) and including the null byte \x00, what were the badchars for OVERFLOW3?

\x00\x11\x40\x5F\xb8\xee

✓ Correct Answer

### TASK 5 : oscp.exe - OVERFLOW4

What is the EIP offset for OVERFLOW4?

2026

✓ Correct Answer

In byte order (e.g. \x00\x01\x02) and including the null byte \x00, what were the badchars for OVERFLOW4?

\x00\xa9\xcd\x04

✓ Correct Answer

TASK 6 : oscp.exe - OVERFLOW5

What is the EIP offset for OVERFLOW5?

314

✓ Correct Answer

In byte order (e.g. \x00\x01\x02) and including the null byte \x00, what were the badchars for OVERFLOW5?

\x00\x16\x2f\xf4\xfd

✓ Correct Answer

TASK 7 : oscp.exe - OVERFLOW6

What is the EIP offset for OVERFLOW6?

1034

✓ Correct Answer

In byte order (e.g. \x00\x01\x02) and including the null byte \x00, what were the badchars for OVERFLOW6?

\x00\x08\x2c\xad

✓ Correct Answer

TASK 8 : oscp.exe - OVERFLOW7

What is the EIP offset for OVERFLOW7?

1306

✓ Correct Answer

In byte order (e.g. \x00\x01\x02) and including the null byte \x00, what were the badchars for OVERFLOW7?

\x00\x8c\xae\xbe\xfb

✓ Correct Answer

TASK 9 : oscp.exe - OVERFLOW8

What is the EIP offset for OVERFLOW8?

1786

✓ Correct Answer

In byte order (e.g. \x00\x01\x02) and including the null byte \x00, what were the badchars for OVERFLOW8?

\x00\x1d\x2e\xc7\xee

✓ Correct Answer

TASK 10 : oscp.exe - OVERFLOW9

What is the EIP offset for OVERFLOW9?

1514

✓ Correct Answer

In byte order (e.g. \x00\x01\x02) and including the null byte \x00, what were the badchars for OVERFLOW9?

\x00\x04\x3e\x3f\xe1

✓ Correct Answer

TASK 11 : oscp.exe - OVERFLOW10

What is the EIP offset for OVERFLOW10?

537

✓ Correct Answer

In byte order (e.g. \x00\x01\x02) and including the null byte \x00, what were the badchars for OVERFLOW10?

\x00\xa0\xad\xbe\xde\xef

✓ Correct Answer

CONCLUSION:

Tryhackme platform Stack based buffer overflow attacks task is successfully completed.